MODULE 4.3 · Multimodal Fusion and Training

Audio-Visual Learning

Free supervision from synchronised video — contrastive SSL, AVHuBERT, AudioCLIP, and temporal alignment

9 min	B3 – Apply	L3 – Intermediate	TH-000007 & TH-000010
Duration	Bloom's	Level	Prerequisite

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Session Overview

Natural synchrony as a training signal — the most abundant free supervisory signal in existence

What You Will Learn

- Why audio and video are naturally paired learning signals
- The McGurk Effect as evidence for human audio-visual fusion
- Self-supervised audio-visual learning methods (AVC, AVS, MAVM)
- Contrastive learning with InfoNCE / NTXent objective
- AVHuBERT: masked audio-visual pre-training
- AudioCLIP: three-way audio-image-text contrastive learning
- Sound source localisation: seeing what you hear
- Temporal alignment challenges and engineering solutions

Concept at a Glance

- Duration: 9 minutes
- Bloom's Level: B3 – Apply
- Difficulty: L3 – Intermediate
- Module: 4.3 Multimodal Fusion and Training
- Prerequisites: Audio Features (TH-000007), Fusion Strategies (TH-000010)
- Next: Training Multimodal Systems (TH-000012)

Learning Objectives

By the end of this session you will be able to:

9 min	B3 – Apply	L3 – Intermediate	TH-000007 & 000010
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1. Explain why audio and video are naturally paired and what 'free supervision' means
2. Describe self-supervised audio-visual pre-training pretext tasks (AVC, AVS, MAVM)
3. Apply the InfoNCE contrastive loss to the audio-visual correspondence problem
4. Compare AVHuBERT and AudioCLIP architectures and their key capabilities
5. Identify downstream tasks enabled by audio-visual pre-training
6. Recognise temporal misalignment challenges and the engineering strategies to mitigate them

SECTION 1 — THE NATURAL PAIRING

Why Audio and Vision Belong Together

The physical world generates audio and visual signals simultaneously — this is free supervision

The Natural Co-Occurrence of Audio and Vision

A dog barking, a guitar playing, a car accelerating — the visual event causes the audio event

The Physics of Free Supervision

The physical world generates audio and visual signals simultaneously from the same source. This causal relationship — visual event produces audio event — provides a virtually unlimited training signal. Every video ever recorded is a labelled audio-visual pair. No human annotation needed.

The McGurk Effect: Perceptual Evidence

Audio 'ba' + Video 'ga' lips = Perceived 'da'

The brain automatically fuses audio and visual speech signals. Even knowing the trick, you cannot suppress the illusion. This proves the human brain has a built-in audio-visual fusion mechanism — AI systems that ignore this miss how the world actually works.

Scale of Free Supervision

YouTube content uploaded: 500+ hours per minute

Video clips available: Billions (essentially unlimited)

Labelling cost: $0 — inherently paired by physics

Signal type: Causal: visual event → audio event

Model quality vs scale: Scales with video data, not annotations

Key insight: The most abundant free supervisory signal ever discovered exists in ordinary internet video. This makes audio-visual learning fundamentally different from supervised learning — you need data, not annotations.

Vision → light → 880000 times faster than sound.
Audio → Sound

SECTION 2 — SELF-SUPERVISED LEARNING

Self-Supervised Audio-Visual Pre-Training

+

text + ? + annotation

Learning without labels: use the synchrony of audio and video as the supervisory signal

I like Cat

Known Known

Three Pretext Tasks for Audio-Visual SSL

Self-supervised learning defines tasks whose labels come from the data itself — no human annotation

1 **Audio-Visual Correspondence (AVC)** Binary classification: does this audio come from the same video clip as this video frame? Positive pairs from the same clip; negative pairs from different clips (automatically generated). The model learns: what sounds go with what visuals.

2 **Temporal Audio-Visual Synchrony (AVS)** Regression task: predict the temporal offset (in seconds) between audio and video streams. Forces the model to learn fine-grained temporal alignment. Harder than AVC — requires understanding the dynamics of both streams.

3 **Masked Audio-Visual Modelling (MAVM)** Mask patches in the visual stream and/or spectral bands in audio. Predict the masked tokens using information from the other modality. This is AVHuBERT's approach — forces cross-modal reconstruction and is extremely data-efficient.

Why these work: All three pretext tasks are constructed automatically from unlabelled video. The model must develop deep audio-visual representations to solve them — these representations transfer to downstream tasks (ASR, VQA, retrieval) with far less labelled data.

InfoNCE: The Contrastive Loss for Audio-Visual Learning

Push matching audio-video pairs together; push non-matching pairs apart — in a shared embedding space

$$\mathcal{L} = -(1/N)\ \Sigma_i \log [\ \exp(\mathrm{sim}(v_i, a_i)/\tau)\ /\ \Sigma_j \exp(\mathrm{sim}(v_i, a_j)/\tau)\ ]$$

v = visual embedding | a = audio embedding | sim = cosine similarity | τ = temperature hyperparameter

Positive pair (v_i, a_i):	Same video clip — audio and visual streams are temporally aligned. The loss pushes their embeddings close together (high cosine similarity).
Negative pairs (v_i, a_j), $j \neq i$:	Different video clips — audio from one clip paired with visual from another. Every other example in the batch is automatically a negative.
Temperature τ:	Controls the sharpness of the distribution. Low τ → sharper focus on hard negatives, but training unstable. Typical values: 0.07–0.1. Learned parameter in CLIP.
Batch size effect:	Larger batch = more negatives per anchor = harder negatives = stronger learning signal. CLIP uses batch size 32,768 across multiple GPUs for this reason.

NTXent is symmetric: Compute loss in both directions: audio-to-visual and visual-to-audio. Sum them. This ensures the shared embedding space is symmetric — you can retrieve audio from visual queries AND visual from audio queries equally well.

SECTION 3 — KEY MODELS

AVHuBERT and AudioCLIP

The two most important audio-visual pre-training models and their complementary approaches

AV-HuBERT: Masked Audio-Visual Pre-Training

Shi et al. (Meta AI, 2022) — applying the masked prediction paradigm to audio-visual speech recognition

Architecture: ResNet visual frontend (lip video frames) + convolutional audio frontend (spectrogram) → feature fusion → Transformer. Randomly masks input features (lip patches or audio spectral bands). Predicts discrete cluster labels (k-means pseudo-labels from offline clustering) for masked positions.

Training Data

1,000 hours of unlabelled video: LRS3 (BBC lip reading) + VoxCeleb2. No transcription labels needed for pre-training — only speaker identity (for contrastive loss variants).

Pre-training Objective

Predict the k-means cluster assignment of masked frames using context from both audio and visual streams. Forces the model to perform cross-modal reconstruction — if audio is masked, use video; if video is masked, use audio.

Key Results

State-of-the-art visual speech recognition (lip reading) using only 30 hours of labelled data. Audio-visual ASR degrades gracefully with noise: −10 dB SNR reduces WER by only 15% vs 250% for audio-only model.

Engineering Insight

Masked cross-modal prediction is extremely data-efficient. 1,000 hours unlabelled + 30 hours labelled outperforms fully supervised models trained on hundreds of labelled hours. The masking forces the model to actually use both modalities.

Application: AV-HuBERT enables lip reading and noise-robust ASR for accessibility — real-time captioning for deaf users in noisy environments, video conferencing enhancement, silent speech interfaces.

AudioCLIP: Three-Way Contrastive Learning

Guzhov et al. (ESC 2021) — extending CLIP's success to a simultaneous audio-image-text contrastive objective

Architecture: Three encoders — CLIP ViT image encoder (pretrained, partially frozen), CLIP text encoder (pretrained), CNN14 audio encoder (trained from scratch). Three pairwise contrastive losses: audio-image, audio-text, image-text. Final loss = average of all three. Trained on AudioSet (2M clips, 527 sound categories).

Zero-Shot Audio Classification

Classify audio events from text descriptions without seeing audio training examples. Text prompt: 'sound of a car engine starting'. AudioCLIP retrieves matching audio clips. No audio class labels required.

Audio-to-Image Retrieval

Find images matching an audio clip. Input: 5-second recording of rain. Output: ranked images of rain, umbrellas, wet roads. Enables audio-based visual search — 'find me images that sound like this'.

Text-to-Audio Retrieval

Find audio clips matching a text query. Input: 'sound of thunder'. Output: ranked audio clips by cosine similarity in shared embedding space. Powers audio asset discovery for content creators.

Cross-Modal Embedding Space

Audio, image, and text all occupy the same 512-dimensional space. You can compute audio-text similarity directly without needing image as an intermediary. Enables emergent cross-modal reasoning not explicitly trained for.

Foundation: AudioCLIP is the foundation for audio-visual search engines, content moderation systems, and music information retrieval. Later extended by ImageBind (Meta, 2023) which aligned 6 modalities via images as the binding modality.

SECTION 4 — TEMPORAL ALIGNMENT

The Core Engineering Challenge

Temporal misalignment corrupts audio-visual learning — SyncNet, filtering, and soft alignment strategies

Temporal Alignment: Problems and Solutions

Video encoding delays, camera cuts, and background audio corrupt the natural audio-visual pairing

Video encoding delay: Audio and video may be off by 100–500 ms due to codec processing. At 16 kHz audio and 25 fps video, even a 1-frame video delay means the audio is already 40 ms ahead of the frame it corresponds to.

→ Sample audio within ±0.5 sec of the video frame timestamp. Train the model to be robust to small temporal offsets (within ±100 ms) via data augmentation.

Camera cuts: Video editing cuts while audio continues. A cut in the middle of a sentence pairs audio of word A with video of a completely different scene. This is a false positive pair.

→ Scene cut detection (PySceneDetect) before creating training pairs. Discard pairs that span a detected scene cut.

Background noise / off-screen sounds: Audio contains sounds from off-screen sources (music, ambient sound) that have no visual correspondence in the frame. Model learns spurious correlations.

→ SyncNet: compute lip-audio sync score using a separate pre-trained lip-sync model. Filter training pairs below sync threshold of 5.0.

Edited / narrated content: Documentary narration, dubbed video, screen recordings — audio and video are deliberately de-coupled. Standard audio-visual correspondence assumption fails entirely.

→ Train/domain classifiers to detect narrated content. Filter by speaker face visibility (dlib face detector + lip motion threshold).

Rule of thumb: Invest 40% of your audio-visual project effort in data filtering. High-quality alignment filters (SyncNet + scene cut detection + face visibility) improve model quality more than architecture changes.

Audio-Visual Models: A Comparison

The evolution from simple correspondence models to universal six-modality embedding systems

Model	Year	Pre-training	Modalities	Key Application
AVC (Arandjelovic)	2017	Correspondence	Audio + Video	Sound source localisation
SyncNet	2016	Lip-sync detection	Audio + Lips	AV sync quality filtering
AudioCLIP	2021	Contrastive (3-way)	Audio+Image+Text	Zero-shot audio classification
AV-HuBERT	2022	Masked prediction	Audio + Lips	Visual speech recognition
ImageBind (Meta)	2023	Contrastive	6 modalities	Universal cross-modal retrieval
VideoPoet (Google)	2024	Token prediction	Audio+Video+Text	Audio-visual generation

Trend: Each year, more modalities are unified into a single embedding space. ImageBind (2023) aligns 6 modalities via images as the binding modality — enabling retrieval between any pair of modalities without direct paired training data between them.

Key Takeaways

The six essential points from this concept

Free supervision from video synchrony Physical causality (visual event → audio event) provides virtually unlimited training signal. Every video is a labelled pair. No annotation cost.

Three SSL pretext tasks Audio-Visual Correspondence (same/different clip), Temporal Synchrony (offset regression), Masked Audio-Visual Modelling (cross-modal reconstruction). All labels derived automatically.

InfoNCE is the dominant loss Push same-clip (positive) pairs together, push different-clip (negative) pairs apart. Larger batch = harder negatives = stronger signal. Temperature τ controls sharpness.

AVHuBERT: masked cross-modal prediction 30 labelled hours + 1,000 unlabelled hours of video → SOTA lip reading. Masking forces genuine cross-modal reasoning.

AudioCLIP: three-way contrastive Audio + Image + Text in one shared space. Enables zero-shot audio classification, audio-to-image retrieval, and text-to-audio search from a single model.

Temporal alignment is the engineering bottleneck SyncNet filtering + scene cut detection + face visibility checks determine data quality more than architecture. 40% of effort should go to data filtering.

Note: Next: AI-MM-IN-TH-000012 — Training Multimodal Systems (3-stage pipeline, objectives, LLaVA-1.5 recipe)

AI-MM-IN-TH-000011 Complete

Audio-Visual Learning

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